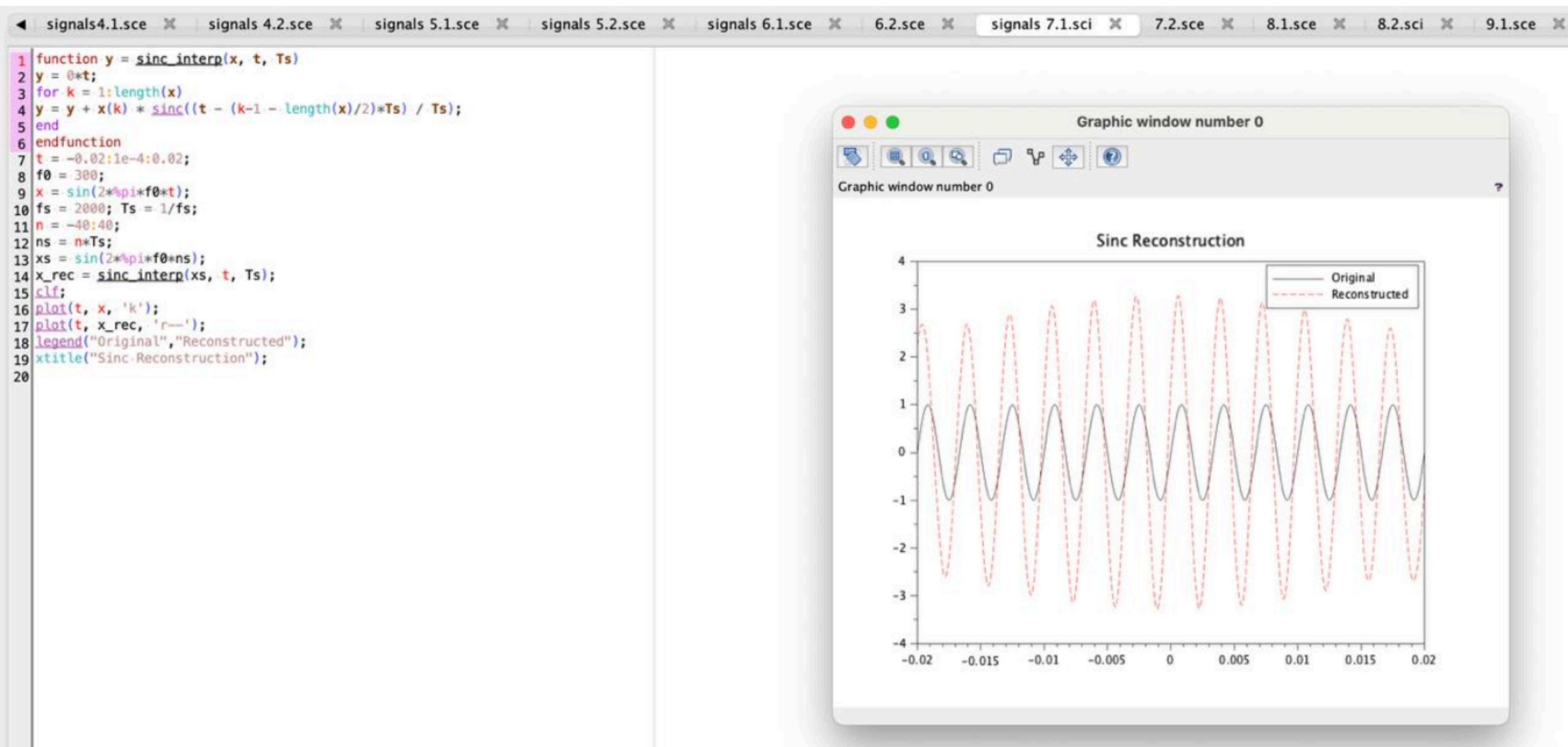




```

1 // Time vector
2 t = -1:0.001:1; // fs = 1000 Hz
3 x = exp(-t.^2); // Gaussian pulse
4 // FFT length (must be >= length(x))
5 N = 2048; // choose bigger than length(x)
6 Xf = fft([x zeros(1, N - length(x))]); // zero-padding to N
7 // Frequency vector
8 fs = 1000; // Hz
9 f = (-N/2:N/2-1) * (fs/N);
10 // Magnitude spectrum
11 magX = abs(fftshift(Xf));
12 // Phase spectrum
13 phaseX = atan(imag(fftshift(Xf)), real(fftshift(Xf)));
14 // Plot
15 clf;
16 subplot(2,1,1);
17 plot(f, magX);
18 xtitle("Magnitude Spectrum", "Frequency (Hz)", "Magnitude");
19 subplot(2,1,2);
20 plot(f, phaseX);
21 xtitle("Phase Spectrum", "Frequency (Hz)", "Phase (rad)");
22

```

